Two nested quadruples

Problèmes ouverts — A. Dujella · 2026

Problem 4.4 (Dujella, 2026). Do there exist two Diophantine quadruples $\{a_1, b, c, d\}$ and $\{a_2, b, c, d\}$ such that $a_1 < a_2 < b < c < d$?

Remark. It was shown by Cipu, Dujella and Fujita (Mediterr. J. Math. **19** (2022), Article 187) that there are only finitely many such examples; in fact, the largest element satisfies $d < 10^{10^{26}}$.